English

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ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025-2026)

Test Pattern

NEET (UG)

MINOR

28-09-2025

PRE-MEDICAL : LEADER & ACHIEVER COURSE (SRG)

Test Booklet Code

E1

This Booklet contains **30** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

Your Target is to secure Good Rank in Pre-Medical 2026

SUBJECT : PHYSICS

Topic : KINEMATICS COMPLETE , NLM (UP TO BB-2), ELECTROSTATICS , CAPACITOR (WITHOUT CIRCUIT PROBLEM).

1. At an instant velocity and acceleration of a particle are $\left(-\hat{i}+\hat{j}+2\hat{k}\right)$ m/s and $\left(3\hat{i}-\hat{j}+\hat{k}\right)$ m/s^2 respectively. At the given instant, speed is -

(1) increasing (2) decreasing

(3) constant (4) can't say

2. A ball is thrown upwards. It returns to ground in a parabolic path. Which of the following is variable.

(1) Speed
(2) Kinetic energy
(3) Vertical component of velocity
(4) Horizontal component of velocity

(1) 1, 2, 3 (2) 1, 2, 4

(3) 2, 3, 4 (4) 1, 3, 4

3. Velocity (in m/s) of a particle moving along a straight line is given by v = $(2t^2 - 3t + 1)$, then match the following :-

	List-1		List-2
(A)	Acceleration (in m/s^2) of particle at t = 1 sec is	(P)	1
(B)	Velocity (in m/s) of particle at t = 2 sec is	(Q)	$\frac{3}{2}$
(C)	Time (in second) when its acceleration is zero	(R)	3
(D)	Magnitude of average acceleration of particle in first 2 seconds in (in m/s^2)	(S)	$\frac{3}{4}$

(1) A ⟶ R ; B ⟶ P ; C ⟶ Q ; D ⟶ S

(2) A ⟶ P ; B ⟶ R ; C ⟶ Q ; D ⟶ S

(3) A ⟶ P ; B ⟶ R ; C ⟶ S ; D ⟶ Q

(4) A ⟶ P ; B ⟶ R ; C ⟶ S ; D ⟶ P

4. **Assertion(A) :** The average speed of an object may be equal to arithmetic mean of individual speeds.

Reason(R) : Average speed is the ratio of total distance travelled and total time taken.

(1) If both assertion and reason are true and reason is the correct explanation of assertion

(2) If both assertion and reason are true but reason is not the correct explanation of assertion

(3) If assertion is true but reason is false

(4) If assertion is false but reason is true

5. Trajectory of a particle in a projectile motion is given by:

$y = x - \frac{x^2}{80}$

Here, x and y are in metres. For this projectile motion, match the entries of column I with the entries of column II.

(Given g = 10m/s^2)

	Column-I		Column-II
(A)	Angle of projection	(P)	20 m
(B)	Angle of velocity with horizontal after 4s	(Q)	80m
(C)	Maximum height	(R)	45°
(D)	Horizontal range	(S)	$\tan^{-1}\left(\frac{1}{2}\right)$

(1) A-R, B-R, C-P, D-Q

(2) A-S, B-R, C-Q, D-P

(3) A-R, B-S, C-P, D-P

(4) A-S, B-S, C-P, D-Q

6. ***Statement-I:*** For a moving object, |displacement| $\leq$ distance.
Statement-II: instantaneous speed is equal to the magnitude of the instantaneous velocity.

(1) Both statement I and statement II are incorrect

(2) Statement I is correct and statement II is incorrect

(3) Statement I is incorrect and statement II is correct

(4) Both statement I and statement II are correct

7. ***Assertion (A) :-*** A projectile that traverses a parabolic path show deviation from its idealised trajectory in the presence of air resistance.
Reason (R) :- Air resistance affect the motion of the projectile.

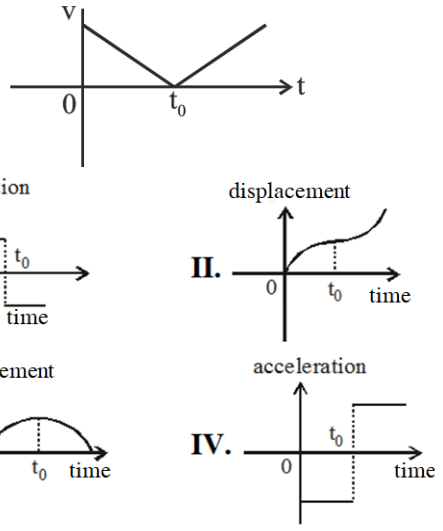
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

8. Velocity v/s time graph of a particle starting from origin is given below. Choose the correct option for corresponding acceleration and displacement graphs :-



(1) II, I (2) III, I

(3) II, IV (4) III, IV

9. Which of the following are correct :-
(a) Average speed of a particle in a given time interval is never less than the magnitude of average velocity.
(b) It is possible to have a situation in which $\left|\frac{d\vec{v}}{dt}\right| \neq 0$ but $\frac{d}{dt}|\vec{v}| = 0$
(c) The average velocity of a particle is zero in a time interval. It is possible than the instantaneous velocity is never zero in the interval
(d) The average velocity of a particle moving on a straight line is zero in a time interval. It is possible that the velocity is never zero in the interval. (Infinite accelerations are not allowed)

(1) a,b,c (2) a,b,d (3) b,c (4) a only

10. A river is flowing from east to west at a speed of 5 m/min. A man on south bank of river, capable of swimming 10 m/min in still water, wants to swim across the river in shortest time; he should swim :-

(1) due north

(2) due north-east

(3) due north-east with double the speed of river

(4) none of the above

11. A lift is moving downward with acceleration 'a'. A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift and a man standing stationary on the ground respectivly :-

(1) g, g (2) g–a, g–a

(3) g–a, g (4) a, g

12. A body A is thrown up vertically from the ground with velocity v_0 and another body B is simultaneously dropped from a height H. They meet at a height $\frac{H}{2}$ if v_0 is equal to.

(1) $\sqrt{2gH}$ (2) $\sqrt{gH}$

(3) $\frac{1}{2}\sqrt{gH}$ (4) $\sqrt{\frac{2g}{H}}$

13. A particle located at x = 0 at time t = 0, starts moving along the positive x-direction with a velocity 'v' that varies as $v = \alpha\sqrt{x}$. The displacement of the particle varies with time as :–

(1) t (2) $t^{1/2}$

(3) t^3 (4) t^2

14. A coin is dropped in a lift. It takes time t_1 to reach the floor when lift is stationary. It takes time t_2 when lift is moving up with constant acceleration, then :-

(1) $t_1 = t_2$

(2) $t_1 > t_2$

(3) $t_2 > t_1$

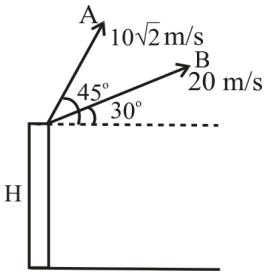
(4) None of these

15. A body falls freely from the top of a tower. It covers 36% of the total height in the last second before striking the ground level. The height of the tower is : (use $g = 10\ ms^{-2}$)

(1) 50 m (2) 75 m

(3) 100 m (4) 125 m

16. Two balls are thrown from a tower as shown in the figure :-



Which of the ball will hit the ground first :-

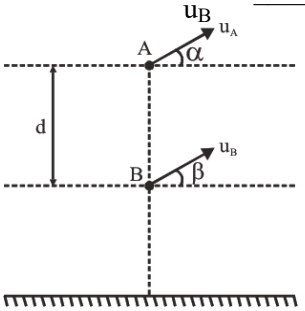
(1) A

(2) B

(3) Both will strike at the same time

(4) Depends on height of tower

17. Two bodies A and B are thrown simultaneously from the different floors of a tower having distance d between them. Velocity of A is u_A at an angle α with horizontal and velocity of B is u_B at an angle β with horizontal. If the two particles collide in mid air $\frac{u_A}{u_B}$ = _____



(1) $\frac{\cos\beta}{\cos\alpha}$ (2) $\frac{\sin\alpha}{\sin\beta}$

(3) $\frac{\tan\alpha}{\tan\beta}$ (4) $\frac{\cot\alpha}{\cot\beta}$

18. Two cities *X* and *Y* are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of 60 km/h in the direction *X* to *Y* notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

(1) 9 min, 40 km/h.

(2) 25 min, 100 km/h.

(3) 10 min, 90 km/h.

(4) 15 min, 120 km/h.

19. If impulse I varies with time t as $I = 20t^2 - 40t$. The change in momentum is minimum at : –

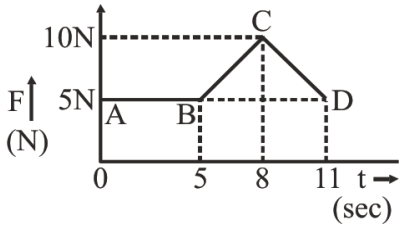
(1) t = 2s

(2) t = 1s

(3) $t = \frac{1}{2}s$

(4) $t = \frac{3}{2}s$

20. A force-time graph for a body is shown. Find the velocity of the body at the end of eleven second. Assume the body to be initially at rest and mass of the body is 10 kg :-



(1) 12 m/s (2) 5 m/s (3) 10 m/s (4) 7 m/s

21. The total external force acting on a rocket is 5000 N in the opposite direction of motion of rocket of mass 1000 kg, if rate of change of mass is 5 kg/s and velocity is 1 km/s then find the acceleration of rocket :

(1) $5\,m/s^2$ (2) $10\,m/s^2$ (3) $0\,m/s^2$ (4) $15\,m/s^2$

22. **A:** Newton's laws of motion are valid at relativistic speed only.
B: Newton's second laws of motion is the fundamental law of motion.
C: Normal reaction may be grater than or less than the weight of a body.
The **correct** statements are :

(1) A, B (2) B, C

(3) Only B (4) All of these

23. Given below are two statement:
Statement-I: Tension in string, Spring force and normal reaction all are conservative forces.
Statement-II: Newton's third law of motion is law of conservation of linear momentum.
In the light of the above statements. Choose the most appropriate answer form the options given below.

(1) Both statement I and statement II are incorrect

(2) Statement I is correct and statement II is incorrect

(3) Statement I is incorrect and statement II is correct

(4) Both statement I and statement II are correct

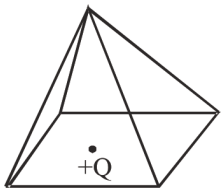
24. Electric field is maximum due to a charged ring of radius R at a distance from circumference of the ring :

(1) $\frac{R}{\sqrt{2}}$ (2) $\sqrt{\frac{3}{2}}R$ (3) $\frac{\sqrt{3}}{2}R$ (4) $\sqrt{2}R$

25. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30º with each other. When suspended in a liquid of density 0.8 g cm^{-3}, the angle remains the same. If density of the material of the sphere is 1.6 g/cm^3 , the dielectric constant of the liquid is:-

(1) 4 (2) 3 (3) 2 (4) 1

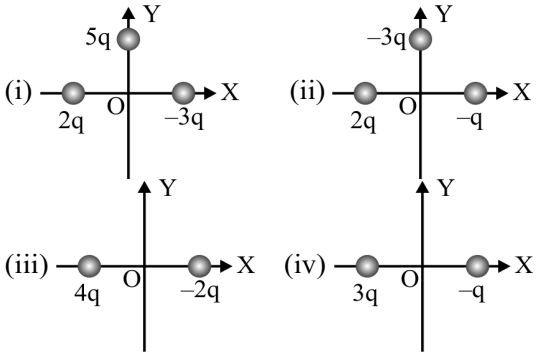
26. A point charge +Q is positioned at the center of the base of a square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is :-



(1) $\frac{Q}{16\varepsilon_0}$ (2) $\frac{Q}{4\varepsilon_0}$

(3) $\frac{Q}{8\varepsilon_0}$ (4) None of these

27. In the following four situations charged particles are at equal distance from the origin. Arrange them the magnitude of the net electric field at origin greatest first :-



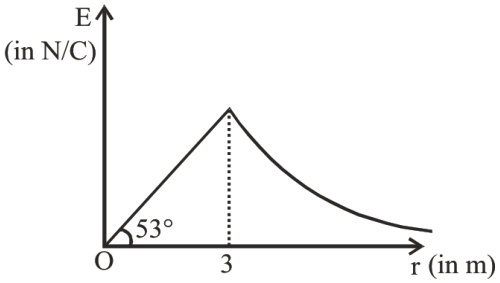
(1) (i) > (ii) > (iii) > (iv)

(2) (ii) > (i) > (iii) > (iv)

(3) (i) > (iii) > (ii) > (iv)

(4) (iv) > (iii) > (ii) > (i)

28. We have a solid non conducting uniformly charged sphere. Its electric field variation with distance from centre is shown in graph then the net charge inside the sphere (in S.I. units) is :-



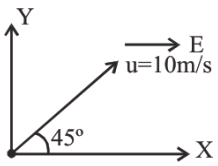
(1) $144\ \pi\in_0$

(2) $100\ \pi\in_0$

(3) $32\ \pi\in_0$

(4) $48\ \pi\in_0$

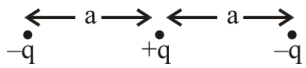
29. A charged particle of mass m = 2 kg and charge 1 μC is thrown from a horizontal ground at an angle of 45° with horizontal with a speed of 10 m/s. In space a horizontal electric field $E = 2 \times 10^7$ N/C exists. Find the range of the projectile. (use = g = 10 ms^{-2})



(1) 5 m (2) 30 m

(3) 10 m (4) 20 m

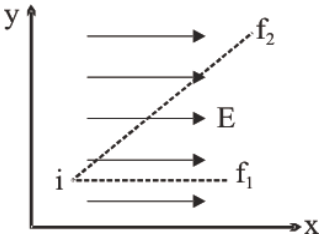
30. Three points charges shown in the figure lie along a straight line. The energy needed to exchange the position of the central +ve charge with one of the negative charges is :-



(1) $\frac{q^2}{8\pi\in_0 a}$ (2) $\frac{3q^2}{8\pi\in_0 a}$

(3) $\frac{q^2}{4\pi\in_0 a}$ (4) $\frac{-q^2}{8\pi\in_0 a}$

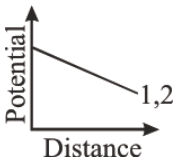
31. A uniform electric field points in the positive x direction, as shown below. Along the two lines f_1, f_2 , we plot the electric potentials as a function of distance. Choose the correct plot.



(1)


(2)

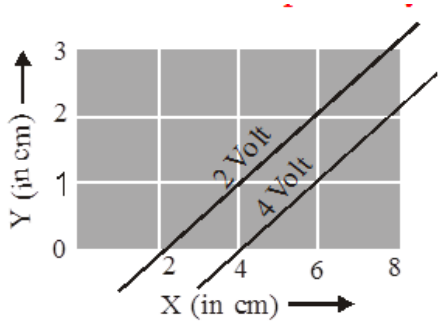

(3)


(4)


32. If potential difference across a capacitor is changed from 10 V to 15 V, then work done by the battery is W. The work done by the battery when potential difference is changed from 15 V to 20 V, will be :

(1) W (2) $\frac{7}{5}$W (3) $\frac{15}{7}$W (4) $\frac{4W}{3}$

33. The figure below shows two equipotential lines in XY plane for an electric field. The scales are marked. The X-component E_x and Y-component E_y of the electric field in the space between these equipotential lines are respectively-



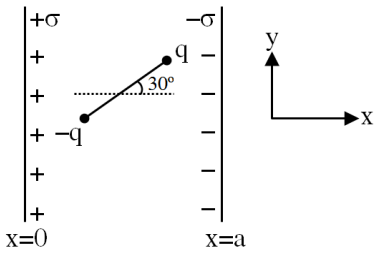
(1) +100 V/m, −200 V/m

(2) +200 V/m, +100 V/m

(3) −100 V/m, +200 V/m

(4) −200 V/m, −100 V/m

34. An electric dipole has charges $-q$ and q separated by d, kept at an angle 30° with x-axis between two large plane sheet of charge as shown in figure. The incorrect option is



(1) Net force on dipole is zero

(2) Torque acting on dipole is $\frac{\sigma qd}{2\epsilon_0}$

(3) Potential energy of dipole is $\frac{-\sigma qd}{2\epsilon_0}\sqrt{3}$

(4) Potential energy of dipole is $\frac{-\sigma qd}{2\epsilon_0}$

35. Three concentric metal shells A, B and C of respective radii a,b and c ($a < b < c$) have surface charge densities $+\sigma$, $-\sigma$ and $+\sigma$ respectively. The potential of shell B is :

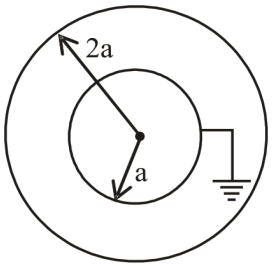
(1) $\frac{\sigma}{\varepsilon_0}\left[\frac{b^2-c^2}{b}+a\right]$

(2) $\frac{\sigma}{\varepsilon_0}\left[\frac{b^2-c^2}{c}+a\right]$

(3) $\frac{\sigma}{\varepsilon_0}\left[\frac{a^2-b^2}{a}+c\right]$

(4) $\frac{\sigma}{\varepsilon_0}\left[\frac{a^2-b^2}{b}+c\right]$

36. In given figure charge given to outer sphere is Q. Inner sphere is grounded. What will be charge on inner sphere after earthing :-



(1) $-\frac{Q}{2}$ (2) Q (3) $\frac{Q}{2}$ (4) $\frac{3Q}{2}$

37. Suppose that earth has a surface charge density of 1 electron/m^2, then calculate Earth's potential.

(1) – 0.115 V (2) – 0.110 V

(3) – 0.105 V (4) – 0.112V

38. An infinite non-conducting sheet has a surface charge density $\sigma = 0.1\ \mu C/m^2$. How far apart are equipotential surfaces whose potentials differ by 50 V ?

(1) 8.8 mm (2) 8.8 cm

(3) 8.8 μm (4) 8.8 pm

39. A particle of mass M and charge q is released from rest in a region of uniform electric field of magnitude E. After a time t, the distance travelled by the charge is S and the kinetic energy attained by the particle is T. Then, the ratio T/S :

(1) remains constant with time t

(2) varies linearly with the mass M of the particle

(3) is independent of the charge q

(4) is independent of the magnitude of the electric field E

40. ***Assertion :-*** The whole charge of a conductor cannot be transferred to another isolated conductor.

Reason :- The total transfer of charge from one to another is not possible.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) Both **(A)** and **(R)** are incorrect

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

41. A sphere of radius 10 cm is charged to a potential of 3000 V. The outward pull per unit area of its surface is :-

(1) 4×10^{-3} N/m^2 (2) 2×10^{-3} N/m^2

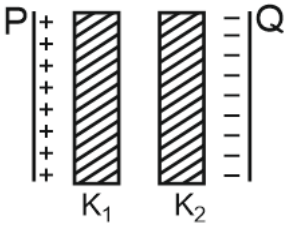
(3) 1×10^{-3} N/m^2 (4) Zero

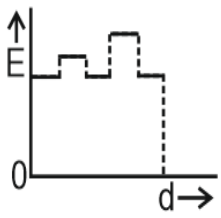
42. A parallel plate capacitor with air between the plates has a capacitance of 9 pF. The separation between its plates is d. The space between the plates is now filled with two dielectrics. One of the dielectrics has dielectrics constant $K_1 = 3$ and thickness d/3 while the other one has dielectric constant $K_2 = 6$ and thickness 2d/3. Capacitance of the capacitor is now :-

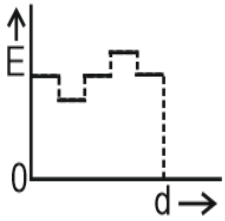
(1) 1.8 pF (2) 45 pF

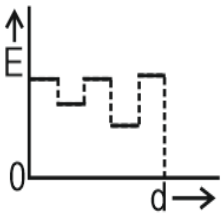
(3) 40.5 pF (4) 20.25 pF

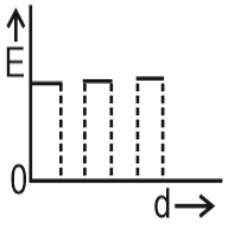
43. Two thin dielectric slabs of dielectric constants K_1 and K_2 ($K_1 < K_2$) are inserted between the plates of a parallel plate capacitor, as shown in the figure. The variation of electric field 'E' between the plates with distance 'd' as measured from plate P is correctly shown by :



(1)


(2)


(3)


(4)


44. A charge of 20μC is placed on the positive plate of an isolated parallel-plate capacitor of capacitance 10μF then potential developed between the plates :-

(1) 1V (2) 2V

(3) 3V (4) 4V

45. Two capacitors with capacity C_1 and C_2, when connected in series, have a capacitance C_S and when connected in parallel have a capacitance C_p. Which of the following is true ?

(1) $C_s = C_1 + C_2$

(2) $C_p = \frac{C_1 C_2}{C_1 + C_2}$

(3) $\frac{C_s}{C_p} = \frac{C_1}{C_2}$

(4) $C_s C_p = C_1 C_2$

SUBJECT : CHEMISTRY

Topic : MOLE CONCEPT , REDOX REACTION, IUPAC , ISOMERISM,GOC (UP TO BB-4) , PERIODIC TABLE , CHEMICAL BONDING (UP TO BB-5) .

46. How many of the following are more acidic than HCOOH.

(i) $CH_2(Cl)-COOH$ (ii) CH_3-COOH

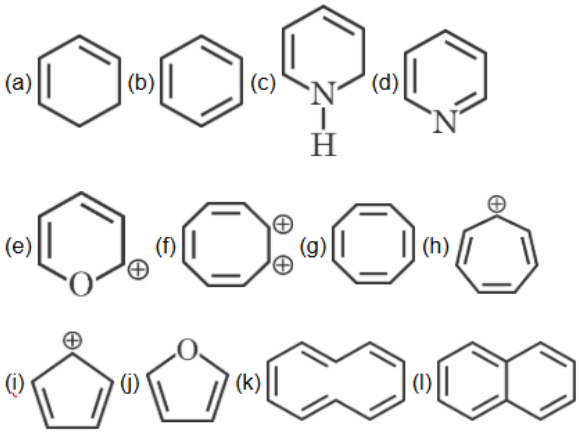
(iii) $\bar{O}OC-CH_2-COOH$ (iv) $NC-CH_2-COOH$

(v) $CH_2(F)-COOH$ (vi) $CH_2(NO_2)-COOH$

(vii) $CH_2(NH^-)-COOH$ (viii) $CH_2(NH_3^+)-COOH$

(1) 3 (2) 4 (3) 5 (4) 6

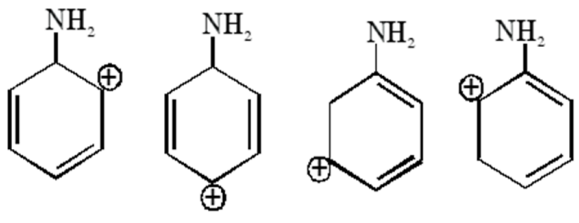
47. How many of the above are aromatic :



(1) 4 (2) 5

(3) 6 (4) 7

48. The most stable carbocation for the following is :



(1) c (2) d

(3) b (4) a

49. Most stable carbocation is formed by the heterolysis of :

(1) $(CH_3)_3CBr$ (2) $(C_6H_5)_3CBr$

(3) $(C_6H_5)_2CHBr$ (4) $C_6H_5CH_2Br$

50. Rank the following free radicals in order of decreasing stability :-

(I) $C_6H_5\dot{C}HC_6H_5$

(II) $C_6H_5-\dot{C}H-CH=CH_2$

(III) $CH_3-\dot{C}H-CH_3$

(IV) $C_6H_5-\dot{C}H-CH_3$

(V) $CH_3CH=CHCH_2\dot{C}H_2$

(VI) $CH_3-CH_2-\dot{C}(CH_3)-CH_3$

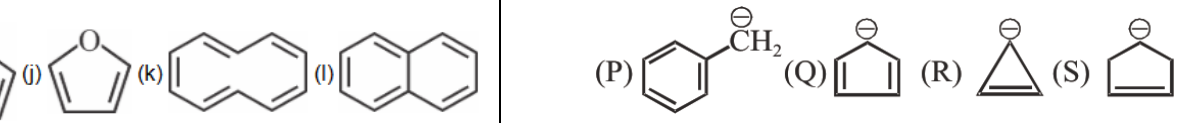
(1) I > II > III > IV > V > VI

(2) VI > V > IV > III > II > I

(3) I > II > IV > VI > III > V

(4) I > IV > VI > V > II > III

51. Stability of carbanion follows the order.



(1) Q > P > S > R (2) Q > P > R > S

(3) P > Q > R > S (4) P > Q > S > R

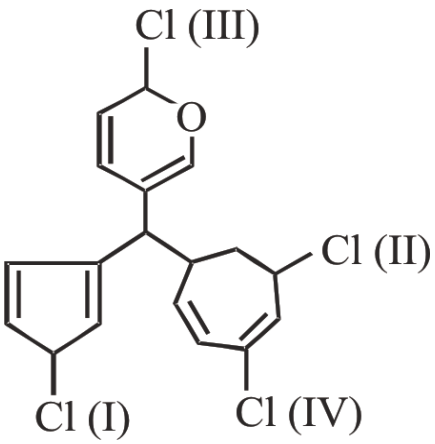
52. Which of the following statement is correct for resonance :-

(1) Canonical form's are hypothetical

(2) All the resonating structure should have same position of atoms

(3) Resonance hybrid is actual structure of molecule

(4) All of these

53. 

Which Cl will react fastest on addition of $AgNO_3$ to form AgCl ?

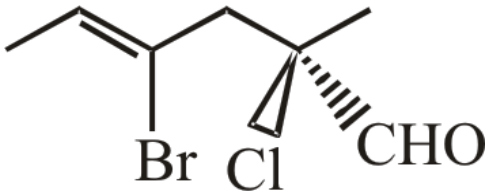
(1) I

(2) II

(3) III

(4) IV

54. The IUPAC name of given compound is :-



(1) (2R, 4E)-4-bromo-2-chloro-2-methylhexanal

(2) (2R, 4Z)-4-bromo-2-chloro-2-methyl-4-hexenal

(3) (2Z, 5R)-3-bromo-5-chloro-5-formyl-2-hexene

(4) (2S, 4E)-4-bromo-2-chloro-2-methyl-4-hexenal

55. Inductive effect involves :

(1) Partial shifting of σ-electrons

(2) delocalization of π electrons

(3) delocatisation of σ electron

(4) displacement of π electrons

56. Match List I with List II.

	List - I (Isomeric pairs)		List - II (Type of isomers)
A.	Propanamine and N-Methylethanamine	I.	Metamers
B.	Hexan-2-one and Hexan-3-one	II.	Positional isomers
C.	Ethanamide and Hydroxyethanimine	III.	Functional isomers
D.	o-nitrophenol and p-nitrophenol	IV.	Tautomers

Choose the correct answer from the options given below :-

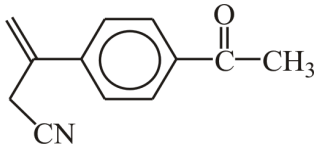
(1) A-III, B-IV, C-I, D-II

(2) A-IV, B-III, C-I, D-II

(3) A-II, B-III, C-I, D-IV

(4) A-III, B-I, C-IV, D-II

57. Double bond equivalent in compound



(1) 6 (2) 7

(3) 8 (4) 5

58. How many different secondary halide names can be assigned to the compounds with molecular formula $C_5H_{11}Cl$ (excluding stereoisomers) ?

(1) 3 (2) 4

(3) 5 (4) 7

59. The total number of possible isomers with molecular formula C_6H_{12} that contains a Cyclobutane ring :-

(1) 5 (2) 6

(3) 7 (4) 8

60. How many total structural isomer of 2° amine are possible with molecular formula $C_4H_{11}N$:

(1) 4 (2) 3 (3) 5 (4) 2

61. Rearrange the following (I to IV) in the order of increasing masses :

(I) 0.5 mole of O_3

(II) 0.5 gram atom of oxygen

(III) 3.011×10^{23} molecules of O_2

(IV) 5.6 litre of CO_2 at STP

(1) II < IV < III < I (2) II < I < IV < III

(3) IV < II < III < I (4) I < II < III < IV

62. Match the column I with column II :

	Column I (Amount of substance)		Column II (Number of oxygen atoms)
(A)	98 mg H_2SO_4	(P)	2 N_A
(B)	18 mL water (d = 1 g/mL)	(Q)	2×10^{-3} N_A
(C)	60 g NO gas	(R)	4×10^{-3} N_A
(D)	44.8 mL of CO_2 gas at 0.5 atm & 273 K	(S)	N_A

Correct match is :

(1) A ⟶ P, B ⟶ Q, C ⟶ R, D ⟶ S

(2) A ⟶ P, B ⟶ S, C ⟶ R, D ⟶ Q

(3) A ⟶ R, B ⟶ S, C ⟶ P, D ⟶ Q

(4) A ⟶ R, B ⟶ S, C ⟶ Q, D ⟶ P

63. Arrange the following in increasing order of masses :

(a) 1 molecule of oxygen

(b) 1 atom of nitrogen

(c) 1 mole of water

(d) 1 g atom of oxygen

(1) a < b < c < d (2) b < a < c < d

(3) b < a < d < c (4) a < b < d < c

64. Phosphine (PH_3) is a poisonous gas which decompose to form molecular phosphorous (P_4) and hydrogen gas. If 200 mL of PH_3 is decomposed then the observed change in volume is : ($4PH_{3(g)} \longrightarrow P_{4(s)} + 6H_{2(g)}$).

(1) 100 mL increase in volume

(2) 100 mL decrease in volume

(3) 200 mL increase in volume

(4) 200 mL decrease in volume

65. 200g impure sample of lime stone on heating produces 22.4L CO_2 at STP. Percentage of pure $CaCO_3$ in the sample of lime stone is :-

(1) 10% (2) 25%

(3) 50% (4) 100%

66. NH_3 is produced according to the following reaction $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$. In an experiment 0.25 mol of NH_3 is formed when 0.5 mol of N_2 is reacted with 0.5 mol of H_2. What is % yield ?

(1) 75% (2) 55%

(3) 33% (4) 25%

67. The ratio of masses of an element A attached with the fixed mass of another element B in their two compounds is 2 : 3. If 48 mass of A is attached with 24 mass of B in first compound then what mass of A is attached with the 12 mass of B in the second compound ?

(1) 72 (2) 36

(3) 24 (4) 12

68. 0.6 g sample of a metal sulphate having formula $M_2(SO_4)_3$ reacts with excess $BaCl_2$ to give 1.2 g $BaSO_4$. What is atomic weight of metal M (Ba = 137, S = 32)

(1) 30.75 (2) 69.7

(3) 55.8 (4) 44.1

69. 2 mole of N_2H_4 loses 16 mole of electron is being converted to a new compound X. Assuming that all of the N appears in the new compound. What is the oxidation state of 'N' in X ?

(1) -1 (2) -2

(3) $+2$ (4) $+4$

70. The pair of compounds that can work both as an oxidising as well as reducing agent is :
(i) SO_2 (ii) H_3PO_3 (iii) NH_3 (iv) HNO_3

(1) (i) and (iii)

(2) (ii) and (iii)

(3) (i) and (ii)

(4) (iii) and (iv)

71. **Assertion :** In a reaction
$Zn_{(s)} + CuSO_{4(aq)} \longrightarrow ZnSO_{4(aq)} + Cu_{(s)}$
Zn is a reductant and itself get oxidized
Reason : In a redox reaction, oxidant is reduced by accepting electrons and reductant is oxidized by losing electron.

(1) Both Assertion and Reason are correct and Reason is a correct explanation for Assertion.

(2) Both Assertion and Reason are correct but Reason is not correct explanation for Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

72. Which of the following statement is true about oxidation state of S in $H_2S_4O_6$?

(1) All S-atoms are in +2.5 state

(2) Two S-atoms are in zero state and other two in +5 state

(3) Two S-atom are in -1 state and other two in +6 state

(4) Both (1) & (2) are correct

73. 6×10^{-3} mole of $KMnO_4$ reacts completely with 9×10^{-3} mole of X^{n+} in acidic medium to give XO_3^- and Mn^{2+}. The value of n will be

(1) 7/3 (2) 10/3

(3) 2/3 (4) 5/3

74. Reaction
$aCr_2O_7^{2-} + bSO_2 + cH^+ \rightarrow xCr^{+3} + ySO_4^{-2} + zH_2O$
the value of $\frac{a+b+c}{x+y+z}$ is –

(1) 2 (2) 1.75

(3) 1 (4) 0.25

75. Sum of oxidation state of oxygen atoms in CrO_5 is :

(1) -10 (2) -2

(3) -3 (4) -6

76. Which of the following order is incorrect ?

(1) Atomic radii – B < Ga < Al < In > Tl

(2) Ionization energy – B > Tl > Ga > Al > In

(3) Electronegativity – F > O > N > C

(4) Electron affinity – S > O > P > N

77. The pair with minimum difference in electronegativity is :-

(1) F, Cl (2) C,H

(3) P, H (4) Na, Cs

78. Which of the following is incorrect :-

(1) O_2^+ is more stable than O_2

(2) O_2 has more number of Antibonding e^- than N_2

(3) C_2 has more number of π bonds than N_2

(4) O_2^+ has more IE than O_2

79. Which of following has maximum number of $d\pi$-$p\pi$ bonding :

(1) CO_2 (2) SO_3 (3) ClO_4^- (4) SO_2

80. Which hybridisation is present in ClO_2 ?

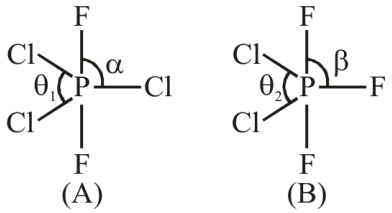
(1) sp

(2) sp^2

(3) sp^3

(4) sp^3d

81. In two given molecules



correct option is :-

(1) θ_1 is equal to θ_2

(2) In A & B both molecules P–Cl bond lengths are longer than axial P–F bond length.

(3) α and β both are equal

(4) Molecule A is nonpolar while molecule B is polar

82. For which of the following species contribution of 's' electron to the shielding constant is 0.30 :-

(a) H^+ (b) He (c) H^-

(d) Li^+ (e) Be^+

Select the correct option :-

(1) a, b, c, d and e

(2) b, c and d only

(3) b, c, d and e only

(4) a, b, c and d only

83. The incorrect order for bond length is :-

(1) $O_2^- > O_2^{-2}$ [O – O]

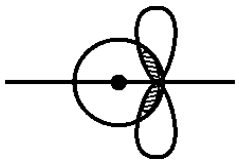
(2) $H_2O_2 > O_2F_2$ [O – O]

(3) $N_2H_4 > N_2F_4$ [N – N]

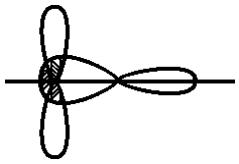
(4) $IF_{7\ (axial)} < IF_{7(equatorial)}$

84. Which of the following is an example of zero overlapping :-

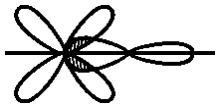
(1)



(2)



(3)



(4) All

85.

I-Element		II-(IE) eV	
(a)	$1s^2 2s^2 2p^6 3s^2$	(i)	10, 120, 135
(b)	$1s^2 2s^2 2p^5$	(ii)	25, 38, 215
(c)	$1s^2 2s^2 2p^6 3s^1$	(iii)	372, 490, 520
(d)	$1s^2 2s^2 2p^6$	(iv)	120, 190, 375

(1) a-ii, b-iv, c-i, d-iii (2) a-i, b-ii, c-iii, d-iv

(3) a-iv, b-iii, c-ii, d-i (4) a-iii, b-iv, c-i, d-ii

86. Correct order of C – O bond length in -

(1) $CO_3^{-2} < CO_2 < CO$

(2) $CO_2 < CO_3^{-2} < CO$

(3) $CO < CO_3^{-2} < CO_2$

(4) $CO < CO_2 < CO_3^{-2}$

87. In which of the following reaction, the hybridisation of the central atom changes from sp^2 to sp^3 ?

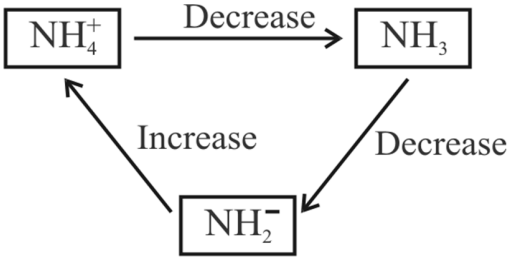
(1) $NH_3 + H^+ \rightarrow NH_4^+$

(2) $AlF_3 + 3F^- \rightarrow AlF_6^{3-}$

(3) $BF_3 + F^- \rightarrow BF_4^-$

(4) $H_2O + H^+ \rightarrow H_3O^+$

88. Which of the following properties show given change in NH_4^+, NH_3 and NH_2^- ?



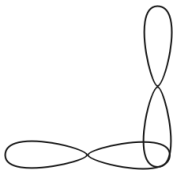
(1) Number of lone pair(s) at nitrogen

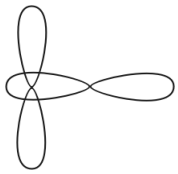
(2) Total number of electrons

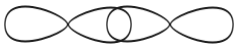
(3) Number of p-orbitals in hybridisation of nitrogen

(4) Bond angle

89. Which p-orbitals overlapping would give the strongest bond ?

(1) 

(2) 

(3) 

(4) 

90. I_3^+ and I_3^- have same:-

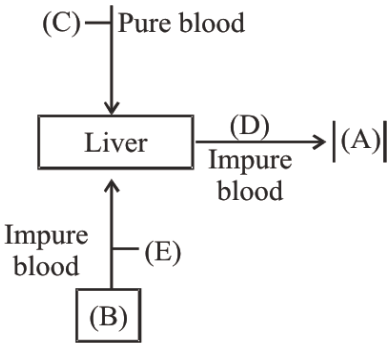
(1) geometry (2) no. of lone pair(s)

(3) Bond angle (4) none of these

SUBJECT : BIOLOGY

Topic : PLANT KINGDOM , PLANT MORPHOLOGY , BREATHING AND EXCHANGE OF GASES, BODY FLUID & CIRCULATION.

91. Please observe the diagram below:-



Here A, B, C, D & E identify correctly :-

(1) A - Hepatic vein, B-Intestine, C-Inferior vena cava

(2) C - Hepatic artery, E-Portal vein, A-Inferior vena cava

(3) D - Hepatic Artery, C-Hepatic Vein, A-Inferior vena cava

(4) B - Intestine, D-Hepatic vein, E-Hepatic artery

92. Given below are two statements :

Statement-I : Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the foetus immediately after the delivery.

Statement-II : Fibrins are formed by the conversion of inactive fibrinogens in the plasma by the enzyme thrombin.

(1) Both **Statement-I** and **Statement-II** are incorrect

(2) **Statement-I** is correct but **Statement-II** is incorrect

(3) **Statement-I** is incorrect but **Statement-II** is correct

(4) Both **Statement-I** and **Statement-II** are correct

93. **Statement-I :** Reduction in number of thrombocytes can cause clotting disorder, leading to excessive loss of blood :

Statement-II : Leucocytes are fragments of megakaryocytes.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct.

94. During ventricular systole –

(1) Oxygenated blood is pumped into the aorta and deoxygenated blood is pumped into the pulmonary artery

(2) Oxygenated blood is pumped into the pulmonary artery and deoxygenated blood is pumped into the artery

(3) Oxygenated blood is pumped into aorta and deoxygenated blood is pumped into pulmonary vein

(4) Oxygenated blood is pumped into pulmonary vein and deoxygenated blood is pumped into pulmonary artery

95. Which of the following statement(s) is/are not correct with respect to platelets ?

(A) They are cell fragments produced from megakaryocytes

(B) Blood normally contains 1,50,000-3,50,000 ml^{-1}

(C) Release a variety of substances most of which are involved in clotting of blood

(D) An increase in their number will lead to excessive loss of blood from the body

(1) A & C (2) B & D

(3) Only D (4) None of these

96. Read the following statement -
(i) Acute chest pain appears.
(ii) The heart stops beating.
(iii) No enough oxygen is reaching the heart muscle.
(iv) Congestion of the lungs is one of the main symptoms of this disease.
(v) It is more common among the middle aged and elderly.
(vi) It occurs due to conditions that affect the blood flow.
How many statement correct about Angina :-

(1) Two (2) Three (3) Four (4) Five

97. Statements :
A. Plasma constitutes 45% of the human blood.
B. Albumin is a plasma protein which helps in osmotic balance
C. Ca^{++} required for blood clotting
D. Plasma without clotting factors is called serum
E. Minerals are not generally found in blood.
Of the above statements :

(1) Only E is wrong and all other A to D are correct.

(2) A and B are correct and C, D and E are wrong

(3) B and D are correct and A, C and E are wrong

(4) B, C and D are correct and A and E are wrong.

98. Which of the following statement(s) regarding the cardiac system is/are correct?
(i) Human heart is an ectodermal in origin.
(ii) Mitral valve guards the opening between the right atrium and left ventricle.
(iii) SAN is located on the left upper corner of the right atrium.
(iv) Stroke volume × Heart rate = Cardiac output

(1) (i) only (2) (i) and (ii)

(3) (ii) and (iii) (4) (iv) only

99. Which of the following statement is incorrect ?

(1) Atria and ventricle of same side are separated by thick muscular tissue

(2) In reptiles left atrium receives oxygenated blood from lungs.

(3) Open circulatory system is present in arthropods

(4) Inter–ventricular septum is thick- walled.

100. Choose the **correct** statements among the following :-

(1) SAN is the pace maker that generates minimum numbers of action potential

(2) During each cardiac cycle, the 'Lub' sound is due to closure of semilunar valves

(3) Cardiac output in each cardiac cycle is approximately 170 ml of blood

(4) The opening between left atrium and left ventricle is guarded by bicuspid valve

101. Which of the following statement incorrect?

(1) The cardiac output of a person varies according to his physiological dema.

(2) End diastolic volume of auricles is about 180 ml.

(3) Pulse pressure of a healthy individual is 40 mmHg

(4) More carbondioxide in blood is called as hypercapnia.

102. What is true about RBCs in human ?

(1) They do not carry CO_2 at all.

(2) They carry about 20-25% of CO_2.

(3) They transport 99.5% of O_2.

(4) They transport about 80% oxygen only and the rest 20% of it is transported in dissolved state in blood Plasma.

103. Which of the following statement is **false** ?

(1) Blood from the right side of the heart is carried to the lungs by the pulmonary artery.

(2) Coronary artery carries oxygenated blood to heart.

(3) Heart beat rate is accelerated by Ach and parasympathetic nerves.

(4) During the ventricular systole, both atria receive the blood.

104. Choose the incorrect answer given below :-

(1) Platelets are produced from megacaryocytes of bone marrow.

(2) Neutrophils are most abundant cells (60–65%) of total WBCs.

(3) Basophils are least (0.5 to 1%) among the WBCs.

(4) A healthy adult has on average 5.5 milion of WBCs.

105. Which of the following statements are correct ?
(i) Ca^{+2} is necessary for blood coagulation
(ii) Coagulation in blood vessel is prevented during normal condition by heparin
(iii) Clotting of blood involves changes of fibrinogen to fibrin by thrombin
(iv) Blood clotting involves cascading process involving a number of factors present always in the active from -

(1) (i), (iii) and (iv) (2) (ii) and (iv)

(3) (i), (ii) and (iii) (4) (iii) and (iv)

106. Read the given statements.
Statement - I : Cardiac arrest is the condition in which heart stops beating.
Statement - II : Heart attack is the condition in which heart muscles is suddenly damaged.

(1) Statement -I^{st} is correct but IInd is incorrect

(2) Statement-IInd is correct but I^{st} is incorrect

(3) Both statement are correct

(4) Both statement are incorrect

107. Consider the following four statements (a-d) and select the option which includes all the correct :-
(a) Coronary Artery disease, (CAD) often referred to as Atherosclerosis
(b) Heart failure means when the heart muscle is suddenly damaged by an inadequate blood supply
(c) High blood pressure leads to heart diseases and also affects vital organs like brain and kidney
(d) Angina occurs due to conditions that affect the blood flow. Create acute chest pain

(1) Statements (b), (c) and (d)

(2) Statements (a), (b)

(3) Statements (c), (d) only

(4) Statement (a), (c) and (d)

108. Which of the following statements is incorrect ?

(1) The RBCs of human are circular disc and Biconcave.

(2) Blood clotting factors are present in plasma.

(3) Human RBCs lack nuclei but have mitochondria.

(4) In adults erythropoiesis occurs in the bone marrow.

109. Read the following statements :-
(a) Systemic circulation takes place between left ventricle and right atrium of heart.
(b) Adrenal medullary hormones can decrease the cardiac output.
(c) Heart failure is not the same as heart attack.
(d) The opening between left atrium and left ventricle is guarded by bicuspid valve.
Out of these how many statements are correct ?

(1) One (2) Two

(3) Three (4) Four

110. Cardiac activity could be moderated by the autonomous neural system. Tick the correct answer:-

(1) The parasympathetic system stimulates heart rate and stroke volume

(2) The sympathetic system stimulates heart rate and stroke volume

(3) The parasympathetic system decreases the heart rate but increases the stroke volume

(4) The sympathetic system decreases the heart rate but increase stroke volume

111. ECG is a graphical representation of the electrical activity of the heart during a cardiac cycle. Mark the correct statement regarding specific electrical activity of the heart.

(1) The contraction of the ventricles start just before S-wave.

(2) The end of the T-wave marks the end of ventricular systole.

(3) QRS complex represents the return of ventricles from excited to normal state.

(4) T-wave represents repolarisation of atria.

112. Choose incorrect statement about heart of human :-

(1) Entire heart made up of cardiac muscles.

(2) Nodal muscular in right upper corner of RA is pacemaker.

(3) Nodal muscular in lower left corner of RA can generate 70-75 min^{-1} action potential.

(4) Nodal musculature is auto-excitable

113. Match the following columns and choose the correct answer from the options given below

Column (I)		Column (II)	
(A)	Erythrocytes	(i)	Most abundant white blood cells and the main phagocytic cell of the blood
(B)	Eosinophils	(ii)	Least abundant white blood cells
(C)	Neutrophils	(iii)	Resist infections and are associated with allergic reactions
(D)	Basophils	(iv)	Blood cells that contain hemoglobin and transport oxygen
(E)	Lymphocytes	(v)	Specialized antibody producing white blood cells

	A	B	C	D	E
(1)	iv	iii	v	i	ii
(2)	i	ii	iii	iv	v
(3)	ii	iii	i	v	iv
(4)	iv	iii	i	ii	v

114. Read the following four statements (A-D) :-

(A) In amphibia and reptiles incomplete double circulation present.

(B) In arthropods and molluscs open circulatory system present.

(C) In joint diastole stage semilunar valves are closed.

(D) The hepatic portal vein carries blood from intestine to the kidney.

How many of the above statements are **wrong**?

(1) Four (2) One

(3) Two (4) Three

115. Given below are four statements (a-d) regarding human blood circulatory system :-
(a) Arteries are thick walled and have narrow lumen as compare to veins.
(b) Angina is acute chest pain when the blood circulation to the brain is reduced.
(c) Vagus secretion acetylcholine result the tachycardia (increased heart rate).
(d) Vertebrates possess Myogenic heart
How many statement are correct from above?

(1) a and b (2) b and c

(3) c and d (4) a and d

116. which statements are correct about normal cardiac cycle-
I - Duration of cardiac cycle is 0.8 sec.
II - Each ventricle pumps out approximately 70 ml of blood/beat.
III - During cardiac cycle 1st heart sound dub is produced.
IV - 1st heart sound is associated with the opening of semilunar values.

(1) I & II are correct

(2) I & III are correct

(3) I, III & IV are correct

(4) Only IVth is correct

117. Which of the following statement is not correct ?

(1) Hyper tension is the term for blood pressure that is higher than normal

(2) Atherosclerosis can be caused by deposits of calcium, fat, cholesterol etc. in the blood vessels that supply blood to the heart muscle

(3) A symptom of acute chest pain due to no enough oxygen to the heart muscles, is cornary artery disease

(4) Beating of heart stops in cardiac arrest

118. What is not incorrect about first heart sound.

(1) Produced at begining of ventricular systole

(2) Sharp sound

(3) Produced due to clossure of semilunar valve

(4) It is Dup sound

119. Which has the thickest walls :-

(1) Right auricle (2) Left auricle

(3) Right ventricles (4) Left ventricle

120. What will happen if a Rh^- person donate blood to a Rh^+ person for the first time

(1) Rh^- person will die

(2) Rh^+ person will die

(3) Nothing will happen to both

(4) Rh^- will live and Rh^+ would die

121. If the stroke volume of heart increases with the cardiac output remaining the same, the heart beats per minute will

(1) decrease (2) remain unaltered

(3) increase (4) become erratic

122. ***Assertion :-*** The pause between the end of the second sound and the beginning of the first sound coincides with ventricular diastole.
Reason :- The second sound (dupp) is created by the closure of the semilunar valve.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

123. **Assertion :** Atherosclerosis affects the vessels which supply blood to the heart muscle.
Reason : Atherosclerosis caused by deposition of calcium fat, cholesterol and fibrous tissue which makes the lumen of arteries narrower.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

124. **Assertion :** Congestion of lungs occurs in heart failure.
Reason : Inefficient pumping of heart causes fluid to build up in lungs.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

125. Match the following columns and select the correct option :

Column - I	Column - II
(a) Pneumotaxic Centre	(i) Alveoli
(b) O_2 Dissociation curve	(ii) Pons region of brain
(c) Carbonic Anhydrase	(iii) Haemoglobin
(d) Primary site of exchange of gases	(iv) R.B.C.

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(3) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(4) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

126. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

(1) High pO_2, low pCO_2, less H^+, lower temperature

(2) Low pO_2, high pCO_2, more H^+, higher temperature

(3) High pO_2, high pCO_2, less H^+, higher temperature

(4) Low pO_2, low pCO_2, more H^+, higher temperature

127. Match the pair :-
a $\longrightarrow$ Hb-O_2 (i) Carboxyhaemoglobin
b $\longrightarrow$ Hb-CO_2 (ii) Oxyhaemoglobin
c $\longrightarrow$ Hb-Fe^{3+} (iii) Carbaminohaemoglobin
d $\longrightarrow$ Hb-CO (iv) Methaemoglobin

(1) a-ii, b-i, c-iii, d-iv (2) a-ii, b-iii, c-i, d-iv

(3) a-ii, b-iii, c-iv, d-i (4) a-i, b-ii, c-iii, d-iv

128. CO_2 is dissociated from carbaminohaemoglobin at alveolar level because of :

(1) P_{CO_2} is low and P_{O_2} is high

(2) Both P_{CO_2} and P_{O_2} are low.

(3) P_{CO_2} is high and P_{O_2} is low.

(4) Both P_{CO_2} and P_{O_2} are high

129. Which of the following statements are true/false ?
(a) The blood transports CO_2, comparatively easily because of its higher solubility.
(b) Approximately 23% of CO_2 is transported being dissolved in the plasma of blood.
(c) The carbondioxide produced by the tissues, diffuses passively into the blood stream and passes into red blood corpuscles and react with water to form H_2CO_3.
(d) The oxyhaemoglobin (HbO_2) of the erythrocytes is basic in nature.
(e) The chloride ions diffuse from plasma into the erythrocytes to maintain ionic balance.

(1) a, c, and e are true and 'b' and 'd' are false

(2) a, c and e are false and 'b' and 'd' are true

(3) a, b and d are true and 'c' and 'e' are false

(4) a, b and d are false and 'c' and 'e' are true

130. Which of the following options correctly represents the lung conditions in asthma and emphysema, respectively ?

(1) Inflammation of bronchioles; Decreased respiratory surface

(2) Increased number of bronchioles; Increased respiratory surface

(3) Increased respiratory surface; Inflammation of bronchioles

(4) Decreased respiratory surface; Inflammation of bronchioles

131. (i) Solubility of gases and thickness of diffusion membrane can affect the rate of diffusion of the gases.

(ii) Every 200 ml of deoxygenated blood delivers approximately 4 ml of CO_2 to the alveoli.

(iii) At the alveolar site where $_PCO_2$ is low, formation of CO_2 & H_2O takes place from H_2CO_3 catalysed by carbonic anhydrase.

(iv) Minute (small) quantity of carbonic anhydrase enzyme is present in RBC.

Identify the option which include **correct statements**.

(1) (ii) & (iii) (2) (i) & (iii) only

(3) (i) & (iv) (4) (i), (iii) & (iv)

132. Read the following statements and choose that how many of them are incorrect-

(A) Pleural fluid reduces friction on lung surface

(B) Terminal bronchiole have incomplete cartilagenous rings

(C) Relaxation of diaphragm increases thoracic volume

(D) Partial pressure of CO_2 in atmospheric air is 0.03 mmHg

(1) One (2) Two

(3) Three (4) Four

133. Read statements given below and choose how many are correct -

(A) Diaphragm is dome shaped

(B) Pneumotaxic centre reduces duration of inspiration

(C) Epiglottis is thin cartilagenous flap over oesophagus

(D) RBCs carry only 20-25% CO_2

(1) One (2) Two (3) Three (4) Four

134. Which factor can cause right shift in oxygen dissociation curve ?

(1) Decreased temperature and increased pH.

(2) Increased temperature and decreased pH.

(3) Increased temperature and increased pH.

(4) Decreased temperature and decreased pH.

135. Choose the correct pair of respiratory volumes and their values :-

(1) Tidal volume = 150 ml

(2) Expiratory reserve volume = 3000 ml

(3) Inspiratory reserve volume = 1000 ml

(4) Residual volume = 1200 ml

136. Which of the following is vascular cryptogam?

(1) *Cedrus* (2) *Equisetum*

(3) *Ginkgo* (4) *Marchantia*

137. Analyse the following characteristics and identify the group of organisms :-

(A)They reproduce sexually by the fusion of non-motile gametes

(B) They show complex developmental changes after fertilization

(C) Majority of organisms are marine with greater concentration found in warmer areas.

(D) They have chlorophyll-a,d and phycoerythrin as major pigments

(1) Phaeophyaceae (2) Chlorophyaceae

(3) Rhodophyaceae (4) Dinoflagellates

138. Read the **statements** (A–D) :

(A) Shrubs are absent in gymnosperm while Herbs are absent in pteridophytes.

(B) Gymnosperms are usually tall tree

(C) Gymnosperms are physiologically Xerophytes

(D) needle like leaves reduce the surface area to minimise water loss in Pinus.

How many of the statement is/are correct ?

(1) Two (2) Three

(3) Four (4) One

139. Sporic meiosis is found in :-

(1) *Riccia* (2) *Marchantia*

(3) *Funaria* (4) All of these

140. In which of the following group there is no independent free living existance of gametophyte ?

(1) Thallophyta (2) Pteridophyta

(3) Bryophyta (4) Gymnosperm

141. How many statements about Bryophytes are correct :

(a) Sterile jacket around reproductive organs

(b) Archegonium like a flaskshaped.

(c) Sporophytic structure totally or partially dependent on gametophyte.

(d) Vascular tissues are well developed

(1) 1 (2) 2 (3) 3 (4) 4

142. Which one of the following statements is **wrong**?

(1) Agar-agar is obtained from *Gelidium* and *Gracilaria*

(2) *Laminaria* and *Sargassum* are used as food

(3) Algae increase the level of dissolved oxygen in the immediate environment

(4) Algin is obtained from red algae, and carrageenan from brown algae.

143. Identify the correct set of statements :

(a) Angiosperms range in size form the smallest Wolfia to tall tree like Eucalyptus. (Over 100 meter)

(b) In angiosperms embryo sac has three celled egg apparatus which includes one egg cell and two synergids.

(c) In angiosperms endosperm is haploid and it represents the female gametophyte.

(d) Double fertilization event is unique to gymnosperms.

(e) In angiosperms zygote develops into embryo and PEN develops into endosperm which provides nourishment to developing embryo.

Choose the correct answer from options given below :

(1) (a), (b), (c) and (e) only

(2) (a), (b) and (e) only

(3) (b) and (e) only

(4) (a), (c) and (e) only

144. **Statement-I :** Prothallus is small but multicellular free living mostly photosynthetic thalloid gametophyte.

Statement-II : Prothallus produces gametes.

(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

(3) Only Statement-I is correct.

(4) Only Statement-II is correct.

145. Which of the following organism will have vegetative cells with cellulosic cell wall that is covered on the outside by a gelatinous coating of algin?

(1) *Volvox* (2) *Sargassum*

(3) *Gracilaria* (4) *Chara*

146. Which one of the following floral character is not represented by floral formula-

(1) Position of ovary

(2) Cohesion of stamen

(3) Adhesion of stamen

(4) Petal aestivation

147. Read all statements carefully and select **incorrect** statement :-

(1) Asexual reproduction in filamentous algae commonly takes place by fragmentation

(2) Gemmae are unicellular structure.

(3) Sporophyte of funaria is more elaborate than that of marchantia

(4) Unicellular rhizoids are present in *Marchantia*.

148. Find out the correct match from the following table :-

	Column-I	Column-II	Column-III
(i)	Tap roots	Formed by the direct elongation of radicle	Sugarcane
(ii)	Fibrous roots	Originate from the base of the stem	Wheat
(iii)	Adventitious roots	Arise from parts of the plant other than the radicle	Sweet potato

(1) i only

(2) i and ii

(3) iii only

(4) ii and iii

149. Match the organism with its type of chloroplast :

	Column-I		Column-II
(a)	*Ulothrix*	(i)	Multicellular jacketed sex organs
(b)	*Chlamydomonas*	(ii)	Filamentous green algae with non-motile gametes
(c)	*Chara*	(iii)	Filamentous green algae with motile gametes
(d)	*Spirogyra*	(iv)	Unicellular green algae

(1) a - iv ; b - iii ; c - i ; d - ii

(2) a - iv ; b - iii ; c - ii ; d - i

(3) a - iii ; b - iv ; c - ii ; d - i

(4) a - iii ; b - iv ; c - i ; d - ii

150. Choose the correct match out of the following-

(1) Phaeophyceae - Fucoxanthin, Starch

(2) Rhodophyceae - Phycocyanin, Laminarin

(3) Chlorophyceae - Chlorophyll, Starch

(4) Cyanophyceae - Phycoerythrin, Floridean Starch

151. Which is the correct match for chlorophyceae members ?

(1) Major pigments – chlorophyll a, b

(2) Stored food – mannitol/laminarin

(3) Cell wall – cellulose and algin

(4) Flagella – 2 lateral

152. Most reduced gametophyte generation is found in :-

(1) Bryophyta (2) Pteridophyta

(3) Gymnosperms (4) Algae

153. Mark the correct statement with respect to the plant given below:-



(1) The root of the plant show symbiotic association with BGA.

(2) It has unbranched stem.

(3) It is heterosporous.

(4) It has seed inside the fruit

154. Identify the algal shown in figure and match. Correct match with the given characters is -



	Name of Algae	Number of flagella in gametes	Stored food
(1)	*Polysiphonia*	Absent	Floridean starch
(2)	*Porphyra*	2-apical	Laminarin & starch
(3)	*Herveyella*	2-lateral	Starch
(4)	*Porphyra*	Absent	Floridean starch

155. Which of the following option is not related with *Cycas*?

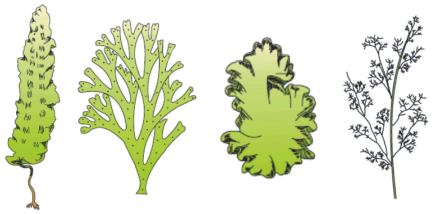
(1) Unbranched stem

(2) Symbiosis with cyanobacteria

(3) More than one Archegonia

(4) Monoecious sporophyte

156. Select correct option about given figures :-



(A) (B) (C) (D)

(1) A and B are red algae

(2) A, B are brown algae and C, D are red algae

(3) A, B, C and D are brown algae

(4) A, B are brown algae and C, D are green algae

157. If the number of chromosomes in root of *Pinus* is 20. What will be the number of chromosomes in its seed coat and endosperm respectively ?

(1) 10 and 20 (2) 30 and 20

(3) 20 and 30 (4) 20 and 10

158. Identify the correct match from the column I, II and III and select the correct option :-

Column-I		Column-II		Column-III	
A	Actinomorphic	i	Bilateral symmetry	a	Mustard
B	Perigynous	ii	Inferior ovary	b	Rose
C	Zygomorphic	iii	Radial symmetry	c	Pea
D	Epigynous	iv	Half-inferior ovary	d	Guava

(1) A-i-a; B-ii-b; C-iii-c; D-iv-d

(2) A-iii-c; B-iv-d; C-i-a; D-ii-b

(3) A-iii-a; B-iv-b; C-i-c; D-ii-d

(4) A-i-c; B-ii-d; C-iii-a; D-iv-b

159. Alternate, pinnately compound/simple stipulated leaf with reticulated venation leguminous roots may be found in :-

(1) *Aloe* (2) Sesbania

(3) Guava (4) Onion

160. Which one of the following is incorrect match regarding family and plant :-

(1) Brassicaceae – *Turnip*

(2) Liliaceae – *Gloriosa*

(3) Fabaceae – *Tulip*

(4) Solanaceae – *Petunia*

161. Placentation in which placenta develops at base of ovary and a single ovale is attached to it. It is known as ...A... & present in ...B... & ...C... family. A, B & C are

(1) A : Basal, B : Asteraceae, C : Liliaceae

(2) A : Basal, B : Asteraceae, C : Poaceae

(3) A : Free central, B : Asteraceae, C : Graminae

(4) A : Marginal, B : Asteraceae, C : Liliaceae

162. **Statement-I :-** Variation in length of stamen filament within a flower is found in cruciferae.
Statement-II :- 5 sepals & 5 petals are found in cruciferae.

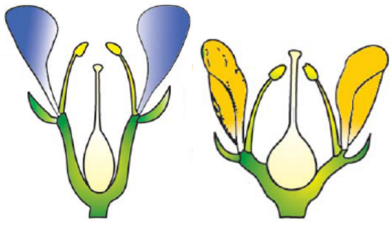
(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is not correct

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and Statement-II are incorrect.

163. How many plants belongs to given diagram :



Mustard, Chilli, Brinjal, Rose, Plum, Peach, Guava, Cucumber, Sunflower.

(1) Three (2) Four (3) Five (4) Six

164. (a) Flower is a modified root
(b) Arrangement of flowers on floral axis is known as inflorescence
(c) In cymose type of inflorescence flowers are arranged in basipetal or acropetal order.
Which of the following statements are not correct ?

(1) a, b, c (2) a, b

(3) b, c (4) a, c

165. What is perianth ?

(1) Outer whorl of flower in which calyx and corolla are not distinguishable seperately.

(2) When stamens and gynoecium are not distinguishable.

(3) When only calyx is present

(4) When calyx and corolla both are absent in a flower.

166. Drupe fruit has

(1) Stony mesocarp (2) Stony endocarp

(3) Stony pericarp (4) Stony epicarp

167. Which one of the following pair is correctly matched ?

(1) Horizontal underground stem – Corm

(2) Aquatic runner – Pistia

(3) Stolon – Pineapple

(4) Phylloclade – Australian acacia

168. A lateral branch with short internodes and each node bearing a rosette of leaves and tuft of roots is found in :

(1) *Nelumbo* and *Pistia*

(2) *Pistia* and *Trapa*

(3) *Pistia* and *Eichhornia*

(4) *Nelumbo* and *Eichharnia*

169. What is common in all plant groups from algae to angiosperm-

(1) Dominant somatic phase is gametophyte

(2) Formation of zygote

(3) Sporophyte as dominant phase

(4) Mitospores

170. In some plants thalamus contributes to fruit formation. Such fruits are termed as :

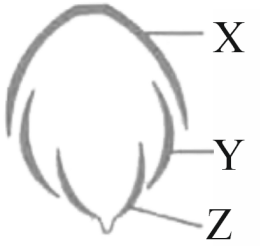
(1) False fruits

(2) Aggregate fruits

(3) True fruits

(4) Parthenocarpic fruit

171. Identify 'X', 'Y' and 'Z' in given diagram :



(1) X - Standard ; Y - Wings ; Z - Keel

(2) X - Wings ; Y - Keel ; Z - Standard

(3) X - Keel ; Y - Wings ; Z - Standard

(4) X - Keel ; Y - Standard ; Z - Wings

172. In Australian acacia, the leaves are small and short-lived, so to perform photosynthesis petioles expand & become flat or sickle shaped. These modified petioles are termed as :-

(1) Cladode

(2) Phylloclade

(3) Phyllode

(4) Pulvinus

173. **Assertion :** In palmately compound leaves rachis is present.

Reason : In palmately compound leaf all the leaflets are attached on common point i.e. at the tip of petiole.

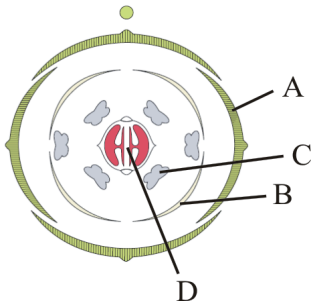
(1) Both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of the **Assertion**.

174. Given below is the diagramatic sketch of floral diagram. Identify the parts labelled A, B, C, D and select the right option about then :-



	Part A	Part B	Part C	Part D
(1)	2+2 arrangement of calyx	Cruciform corolla	Tetradynamous stamen	False septum/ replum
(2)	Cruciform corolla	2+2 arrangement of calyx	False septum/ replum	Tetradynamous stamen
(3)	False septum replum	2+2 arrangement of calyx	Cruciform corolla	Tetradynamous stamen
(4)	Tetradynamous stamen	Cruciform corolla	2+2 arrangement of calyx	False septum/ replum

(1) 1 (2) 2

(3) 3 (4) 4

175. Match the column-I with column-II :-

	Column-I		Column-II
(a)	Fabaceae	(i)	$\oplus \male\!\!\female K_{2+2}C_4A_{2+4}\underline{G}_{(2)}$
(b)	Solanaceae	(ii)	$\oplus \male\!\!\female K_{(5)}C_{1+2+(2)}A_{1+(9)}\underline{G}_1$
(c)	Liliaceae	(iii)	$\oplus \male\!\!\female K_{(5)}\overset{\frown}{C}_{(5)}A_5\underline{G}_{(2)}$
(d)	Brassicaceae	(iv)	$Br \oplus \male\!\!\female \overset{\frown}{P}_{(3+3)}A_{3+3}\underline{G}_{(3)}$

Select the correct answer from the options given below :

	(a)	(b)	(c)	(d)
1	iii	iv	ii	i
2	i	ii	iii	iv
3	ii	iii	iv	i
4	iv	ii	i	iii

176. (a) Primary protonema
(b) Zygote
(c) Leafy gametophyte
(d) Capsule
(e) Spore
Arrange above given structure in order of the life cycle of a moss.

(1) c ⟶ d ⟶ e ⟶ a ⟶ b

(2) c ⟶ b ⟶ d ⟶ a ⟶ e

(3) c ⟶ b ⟶ d ⟶ e ⟶ a

(4) e ⟶ a ⟶ b ⟶ d ⟶ c

177. Which character is related with stamen in Fabaceae and Solanaceae respectively -

(1) Diadelphous and Monothecous

(2) Epiphyllous and Monoadelphous

(3) Dithecous and Polyadelphous

(4) Diadelphous and Dithecous

178. Find the correct statements :-
(a) The dot above the floral diagram represents the mother axis
(b) The bracket represents the "cohesion" in floral formula
(c) Floral formula can show both cohesion and adhesion of stamens

(1) a, b, c

(2) a, c only

(3) b, c only

(4) a, b only

179. Berry is a fruit which is commonly :-

(1) Dry and multiseeded

(2) Dry and single seeded

(3) Fleshy and multiseeded

(4) Fleshy and seedless

180. How many plants in the list given below have axile placentation ?
Mustard, Gram, Tulip, Asparagus, Chilli, Tomato, Arhar, Sunhemp, Colchicum, Onion, Moong, Pea, Tobacco, Lupine

(1) Four

(2) Three

(3) Seven

(4) Ten

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